



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CROP SCIENCE
PAPER 1 Multiple Choice

6049/1

NOVEMBER 2018 SESSION

1 hour

Additional materials:

Multiple Choice answer sheet,
Soft clean eraser,
Soft pencil (type B or HB is recommended),
Electronic calculator.

TIME 1 hour

INSTRUCTIONS TO CANDIDATES

Do not open this booklet until you are told to do so.

Write your name, centre number and candidate number on the answer sheet in the spaces provided unless this has already been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the separate answer sheet.

Read very carefully the instructions on the answer sheet.

INFORMATION FOR CANDIDATES

Each correct answer will score one mark.

Any rough working should be done in this booklet.

This question paper consists of 11 printed pages and 1 blank page.

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Which of the following organelles of the plant cell is rich in RNA?

- A Nucleus.
- B Ribosomes.
- C Lysosomes.
- D Golgi apparatus.

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_____ is responsible for the breakdown of unwanted structures?

- A Mitochondria
- B Ribosome
- C Golgi apparatus
- D Lysosome

3 Which of the following is an effect of organic matter on soil bulk density?

- A Aerates the soil.
- B Decreases soil bulk density.
- C Decrease the volume of soil.
- D Encourages flocculation.

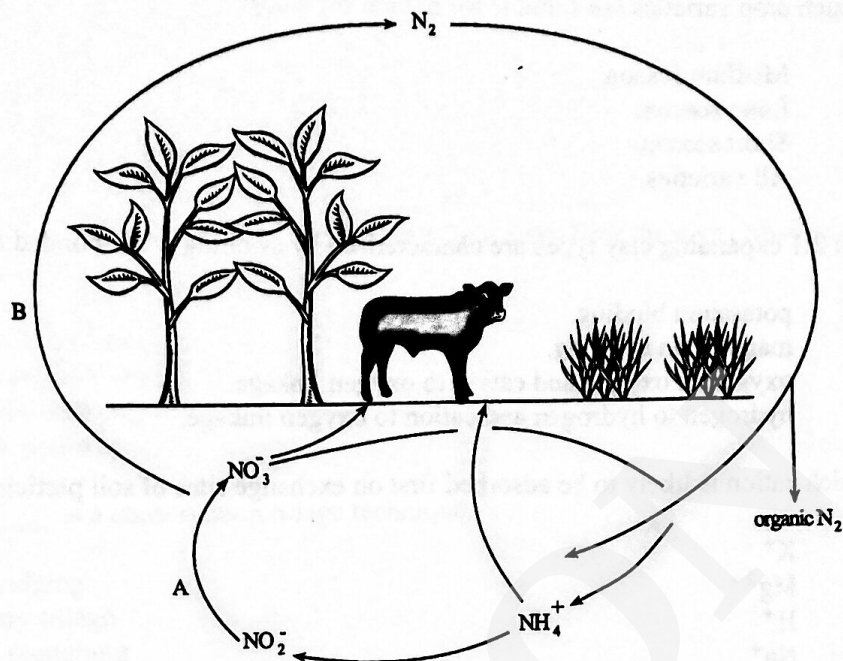
4 Which of the following nutrients is easily leached from the soil?

- A Magnesium.
- B Potassium.
- C Phosphorus.
- D Aluminium.

5 Seed coat rupture during seed germination is due to

- A mechanical scarification.
- B massive imbibition of water.
- C increase in metabolic reactions.
- D massive glycolysis in endosperm.

- 6 Use the diagram below to answer the question that follows.



Which condition is favourable for bacteria at B?

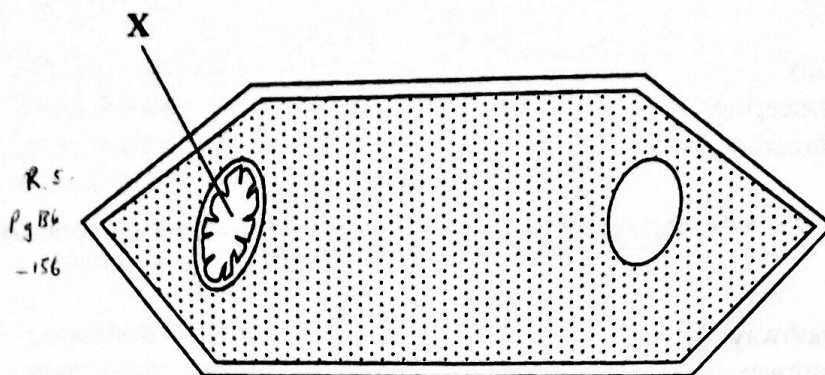
- A Crumb structure.
 - B Waterlogged soils.
 - C Well aerated soils.
 - D Where legumes have been grown.
- 7 The amount of heat required to raise the temperature of 1 kg of water by 1 °C is known as
- A heat of fusion.
 - B thermal capacity.
 - C heat evaporation.
 - D specific heat capacity.
- 8 Which part supplies nutrients to the developing embryo sac during ovule development?
- A Nucellus.
 - B Funiculus.
 - C Integuments.
 - D Ovarian locule.
- 9 Viviparity is common in
- A locusts.
 - B aphids.
 - C weevils.
 - D maize stalk borers.

- 10 Which crop varieties are suitable for natural region II?
- A Medium season.
 - B Long season.
 - C Short season.
 - D All varieties.
- 11 The 2:1 expanding clay types are characterised by swelling when bonded by
- A potassium binding.
 - B magnesium bonding.
 - C oxygen to oxygen and cation to oxygen linkage.
 - D hydrogen to hydrogen and cation to oxygen linkage.
- 12 Which cation is likely to be adsorbed first on exchange sites of soil particles?
- A K^+
 - B Mg^{2+}
 - C H^+
 - D Na^+
- 13 Select the correct source of genetic variation in plants.
- A Pure selection.
 - B Mass selection.
 - C Mutation.
 - D Selfing.
- 14 A case where a phenotypic relation of the heterozygote is greater than that of either homozygous parents is
- A partial dominance.
 - B overdominance.
 - C co-dominance.
 - D epistasis.
- 15 In which reactions are two molecules of CO_2 given off to produce ATP during respiration?
- A Glycolysis
 - B Krebs cycle
 - C Glucose oxidation
 - D Electron transport pathway
- 16 Which of the following growth regulators is associated with abscission in plants?
- A Auxins.
 - B Naphthalene.
 - C Cytokinins.
 - D Gibberelins.

- 17 The transfer of genes between organisms is called
- A genomics.
 - B micro biology.
 - C genetic engineering.
 - D molecular breeding.
- 18 Which pathway draws water through the cell walls in a mass flow through cohesion of water molecules?
- A Symplast pathway.
 - B Apoplast pathway.
 - C Vascular pathway.
 - D Xylem pathway.
- 19 _____ is a conservation tillage technique.
- A Tied ridging
 - B Primary tillage
 - C Deep ploughing
 - D Conventional tillage
- 20 Which of the following factors affects spray drift?
- A Temperature.
 - B Light intensity.
 - C Operational speed.
 - D Chemical composition.
- 21 The cementing substance between two adjacent cells, made of peptic compounds is
- A plasma membrane.
 - B middle lamella.
 - C primary cell wall.
 - D secondary cell wall.
- 22 Which of the following mechanisms contributes to genetic variation in plants?
- A Duplication of chromosomes.
 - B Separation of sister chromatids.
 - C Independent assortment of chromatids.
 - D DNA replication before nucleus division.
- 23 What name is given to the suppression of lateral root emergence or lateral bud growth by an actively growing root or shoot apex?
- A Bud dormancy.
 - B Apical dormancy.
 - C Apical dominance.
 - D Lateral dominance.

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Identify the cell organelle labelled X in the diagram.



- A Chloroplast
- B Mitochondria
- C Golgi apparatus
- D Rough endoplasmic reticulum

25 The result of a higher water loss through the leaves coupled with low water uptake in the roots is increased

- A wilting.
- B respiration.
- C translocation.
- D photosynthesis.

26 In which type of germination are cotyledons lifted above the soil?

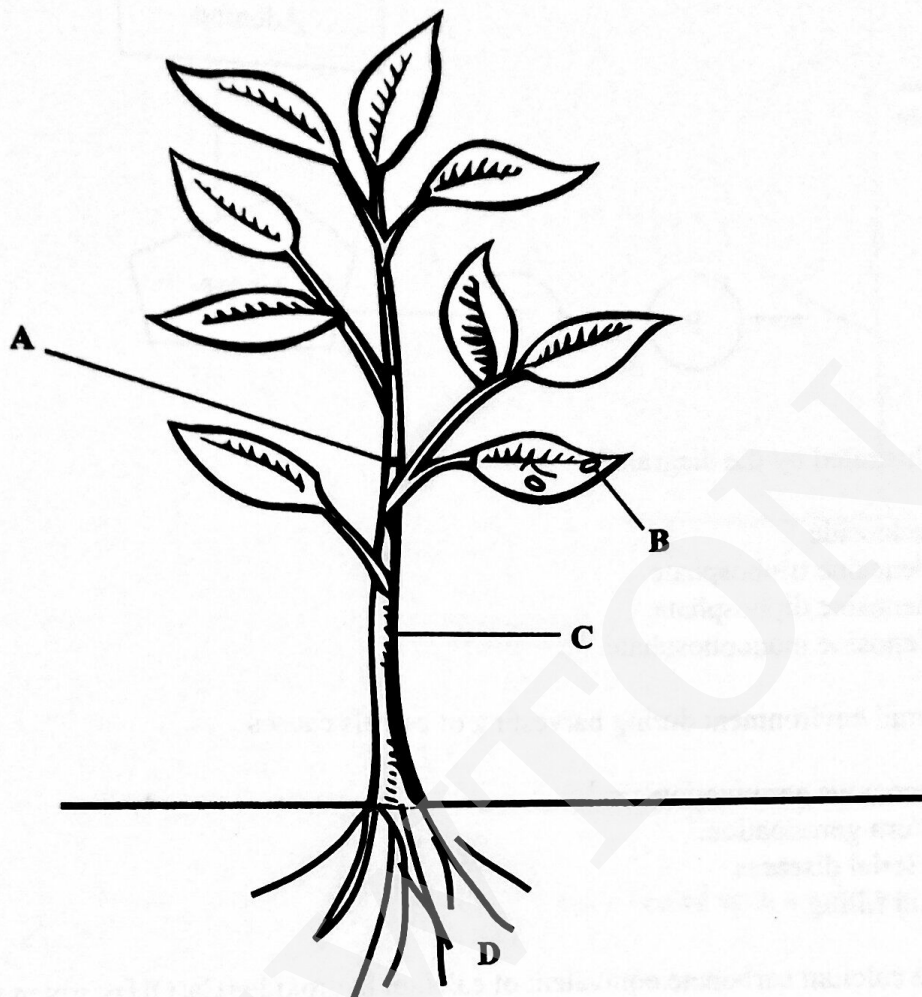
- A Hypocotyl.
- B Emergence.
- C Hypogeal.
- D Epigeal.

27 What is water potential?

- A Movement of molecules within a gradient.
- B Movement of molecules within a substance without a change in temperature.
- C Movement of water from a region of high concentration to a region of low concentration.
- D The difference in free energy per unit volume between metrically bound, pressurised, osmotically constructed water and pure free water.

28 The addition of soluble salts to the root zone when there is insufficient moisture causes

- A temporary wilting.
- B permanent wilting.
- C physiological drought.
- D turgidity.



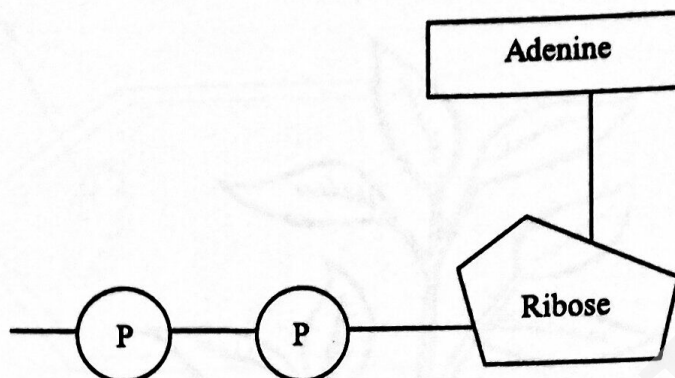
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Which process is indicated by the equation below?



- A Nitrogen fixation by soil colloids
- B Immobilisation of organic matter
- C Mineralisation of organic matter
- D Nitrogen fixation from the atmosphere

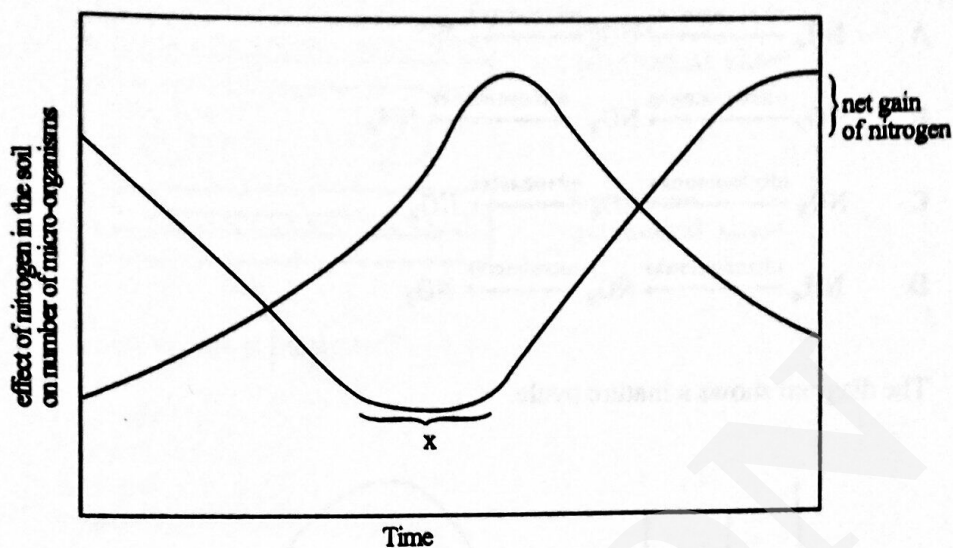
- 31 Use the diagram to answer the question that follows.



What is illustrated by the diagram?

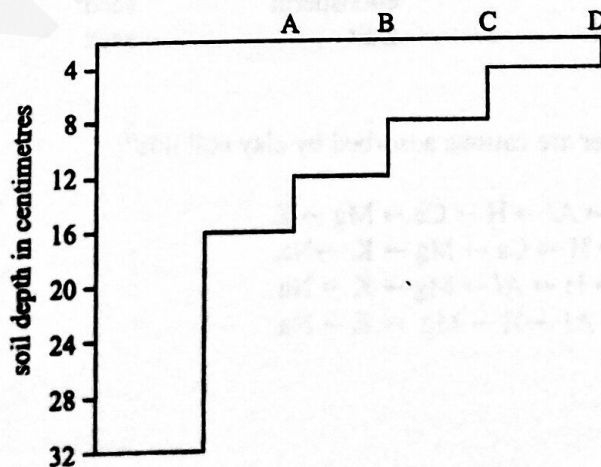
- A Nucleoside.
 - B Adenosine triphosphate.
 - C Adenosine diphosphate.
 - D Adenosine monophosphate.
- 32 A high humid environment during harvesting of cereals causes
- A precocious germination.
 - B pattern germination.
 - C bacterial diseases.
 - D grain filling.
- 33 What is the calcium carbonate equivalent of calcium hydroxide ($\text{Ca}(\text{OH})_2$), given that the molecular weight of calcium = 40 g, Oxygen = 16 g, carbon = 12 g, Hydrogen = 1g.
- A 100 %
 - B 115 %
 - C 125 %
 - D 135 %

- 34 The diagram illustrates the decomposition of organic matter.



What is X?

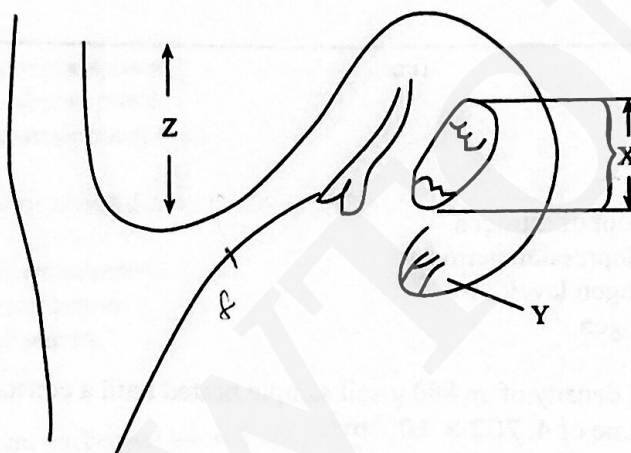
- A Total amount of nitrogen.
 - B Nitrogen depression period.
 - C Initial nitrogen level.
 - D Cross nitrogen.
- 35 Calculate the bulk density of an 800 g soil sample heated until a constant mass of 740 g was achieved at a volume of $4.702 \times 10^{-4} \text{ m}^3$.
- A $1\,701 \text{ kg m}^{-3}$
 - B $1\,573 \text{ kg m}^{-3}$
 - C $1\,27\,605 \text{ kg m}^{-3}$
 - D 157.3 kg m^{-3}
- 36 Where are more bacteria found in the diagram?



37 Which equation illustrates the process of nitrification?

- A $\text{NH}_4 \xrightarrow{\text{nitrosomonas}} \text{NO}_2 \xrightarrow{\text{nitrobacter}} \text{NO}_3$
- B $\text{NO}_2 \xrightarrow{\text{nitrosomonas}} \text{NO}_3 \xrightarrow{\text{nitrosomonas}} \text{NH}_4$
- C $\text{NO}_3 \xrightarrow{\text{nitrosomonas}} \text{NH}_4 \xrightarrow{\text{nitrobacter}} \text{NO}_2$
- D $\text{NH}_4 \xrightarrow{\text{nitrosomonas}} \text{NO}_3 \xrightarrow{\text{nitrobacter}} \text{NO}_3$

38 The diagram shows a mature ovule.



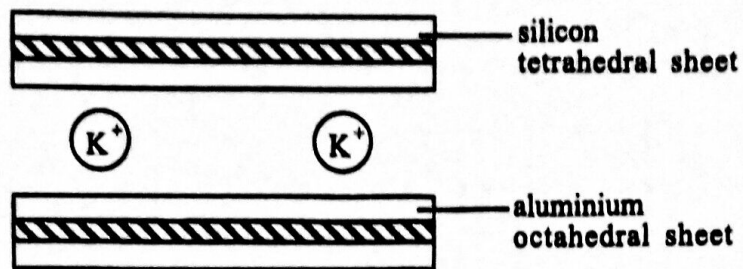
Name the parts labelled X, Y and Z.

	X	Y	Z
A	endosperm	testa	fruit
B	seed	testa	ovary.
C	testa	endosperm	seed
D	testa	fruit	seed

39 In which order are cations adsorbed by clay colloids?

- A $\text{Na} \rightarrow \text{Al} \rightarrow \text{H} \rightarrow \text{Ca} \rightarrow \text{Mg} \rightarrow \text{K}$
- B $\text{Al} \rightarrow \text{H} \rightarrow \text{Ca} \rightarrow \text{Mg} \rightarrow \text{K} \rightarrow \text{Na}$
- C $\text{Ca} \rightarrow \text{H} \rightarrow \text{Al} \rightarrow \text{Mg} \rightarrow \text{K} \rightarrow \text{Na}$
- D $\text{Ca} \rightarrow \text{Al} \rightarrow \text{H} \rightarrow \text{Mg} \rightarrow \text{K} \rightarrow \text{Na}$

- 40 The diagram shows a structure of an aluminosilicate clay.



Which aluminosilicate is illustrated?

- A Kaolinite.
- B Chlorite.
- C Smectite.
- D Mica.